

ProbOS — Month 4 Pre-Plan

Dashboard Architecture and Core Build

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Month goal

Build the core dashboard for ProbOS’s two already-validated domains (battery, pharma): a working, genuinely usable interface for running simulations, configuring priors, and viewing uncertainty results – not a prototype, but the foundation every later domain will build on.

Week 14 – Framework decision + spike

- Spike BOTH candidate frameworks before committing: Streamlit (fast, Python-native, weaker on custom UI control) and a React frontend against the existing FastAPI backend (more “real product” feel, more implementation work).
- Exit criterion: a working “hello world” simulation trigger in BOTH candidates, compared honestly on development speed, customization ceiling, and fit with ProbOS’s existing FastAPI service layer.

Study materials for this week:

- Streamlit – official docs (docs.streamlit.io) are genuinely sufficient; no book needed given this framework’s scope.
- React – react.dev’s official docs (hooks-first, rewritten 2023) over any book; supplement with *Learning React* (O’Reilly, Banks & Porcello) only if the React path is chosen.

Week 15 – Core dashboard build

- Model selector (battery, pharma)
- Parameter/prior configuration UI – must expose real prior distributions (Normal/LogNormal/Uniform), not just point estimates
- Run-simulation trigger, wired to the existing FastAPI `/simulate` endpoint

Week 16 – Visualization layer

- P05/P50/P95 percentile trajectory bands
- Sobol sensitivity bar chart
- Provenance trace explorer for a selected tail-risk outcome

Study materials for this week:

- Cole Nussbaumer Knaflic, *Storytelling with Data* (Wiley, 2015) – the standard reference for designing numeric/ statistical dashboards that communicate uncertainty clearly.
- Edward Tufte, *The Visual Display of Quantitative Information* – foundational visual-design taste before building charts representing safety-critical uncertainty.

Week 17 – Error handling + first real user-flow test

- Loading states, error messages for invalid prior configurations, graceful failure for out-of-range parameters
- First end-to-end user-flow test with someone other than the builder, if possible

Exit criteria

A working dashboard covering both validated domains, with correctly visualized uncertainty, that a cold-start user can operate without explanation.